

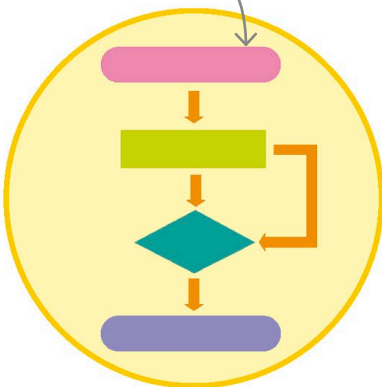
IN DEPTH

Thinking skills



Although computational thinking developed as a way for computer scientists to solve problems, it is useful in many areas. At work, people use decomposition – a key skill in computational thinking – by breaking down jobs into smaller tasks. Sometimes, it's possible to reuse a plan from a previous task for a new one, if there is a similar pattern to the work.

An algorithm can be written as a flowchart.

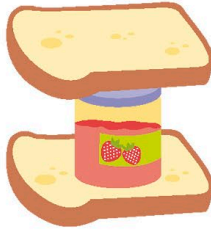


4. Algorithms

An algorithm is a series of instructions that solve a problem. Each instruction must be precise and unambiguous so that there's no doubt about what it means. Programs are algorithms translated into programming languages.

How to think computationally

A good way to practise computational thinking is to step away from the computer. Doing things in a particular order, making decisions, and repeating actions are all elements of everyday tasks. They are also the basic elements of computer programs. Writing instructions for a task that everyone does without much thought can be trickier than expected.



< Precise instructions

Writing a set of instructions accurate enough for a computer to make a sandwich can be quite a challenge. Unlike people, computers can't cope with any ambiguity. To make a jam sandwich it's important to write instructions that can be followed exactly – so the jam ends up inside the sandwich, but not the jam jar!

> Different solutions

Often, particularly in maths and science, kids learn that there's a single right answer to questions. Computational thinking is applied to open-ended problems – those that have many possible solutions.

What would you like for dessert?



< Remixing

Another activity that promotes computational thinking is remixing – taking code that someone has shared and changing parts of it. This allows new coders to build up their understanding gradually and create something new in the process.

> Getting social

Programming is often viewed as an activity carried out in isolation. However, in the workplace, coders usually work in pairs or teams. In addition to online coding communities like Scratch, there are various coding clubs that young people interested in coding and technology can join.

